Mud Pump Abnormal Vibration

Field Manual — Setup and Troubleshooting Reference

Use case definition. A triplex mud pump supplies drilling fluid at controlled pressure and flow. Abnormal behavior includes pulsation, vibration, pressure loss, suction starvation, valve/liner wear, dampener failure, bearing heat, drive overload, and loose piping or mounting. Troubleshooting protects the fluid end, power end, drive, hoses, and drilling process.
Manual setup and use. Use the pump curve, liner size, dampener charge specification, approved pressure rating, and OEM/site alarm or trip limits. Values are training/reference values.

Safety note. Keep clear of pressurized mud lines and pulsation dampeners; isolate, bleed down, and verify zero pressure before maintenance.

Quick-use workflow

Use this sequence for every event: establish a safe operating state; capture the initial numerical readings; compare each value with the stated reference; apply the trigger-based action; correct the identified component or process path; restore operation gradually; and confirm the expected numerical result. If an OEM/site alarm or trip limit differs from a training value, the OEM/site limit governs.

Scenario 1: High vibration

Field Condition

vibration 10.2 mm/s RMS from 3–5; discharge 180 barg; 95 strokes/min; bearing 78 °C

Problem

High vibration indicates fluid-end, power-end, mounting, or piping distress.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses vibration above 8 mm/s RMS.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when vibration 3–5 mm/s RMS.

Expected Result

vibration 3–5 mm/s RMS

Scenario 2: Pulsation

Field Condition

discharge pressure oscillates 28 barg peak-to-peak every 1.2 s; dampener 62 barg

Problem

Failed or incorrectly charged pulsation control is causing pressure cycling.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses pulsation above 10 barg peak-to-peak.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.

5. Restore operation in a controlled ramp and accept the equipment only when pulsation below 5 barg peak-to-peak.

Expected Result

pulsation below 5 barg peak-to-peak

Scenario 3: Suction restriction

Field Condition

suction 0.4 barg from 1.2–1.8; flow 1.8 m^3/min from 2.4; vibration 8.0

Problem

Suction-side restriction is starving the fluid end.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses suction below 0.8 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when suction 1.2–1.8 barg and flow 2.3–2.5 m^3/min.

Expected Result

suction 1.2–1.8 barg and flow 2.3–2.5 m^3/min

Scenario 4: Cavitation

Field Condition

suction pressure falls to −0.15 barg; vibration 9.4; discharge 165 barg

Problem

Negative suction pressure indicates inadequate supply/NPSH.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses suction below 0 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when positive suction 1.2–1.8 barg and vibration 3–5.

Expected Result

positive suction 1.2–1.8 barg and vibration 3–5

Scenario 5: Valve damage

Field Condition

discharge 172 barg; suction 1.4; flow 2.0 m^3/min; valve-chamber temperature 68 °C from 48

Problem

A damaged suction/discharge valve reduces flow and creates local heat.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses temperature rise above 15 °C.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when flow 2.3–2.5 m^3/min and temperature near 45–55 °C.

Expected Result

flow 2.3–2.5 m^3/min and temperature near 45–55 °C

Scenario 6: Liner wear

Field Condition

flow 1.9 m^3/min at 95 strokes/min versus 2.4; discharge 160 barg; liner leakage 3.2 L/min

Problem

Worn liner/piston clearance is reducing volumetric efficiency.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses leakage above 1 L/min.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when leakage below 0.5 L/min and flow 2.3–2.5.

Expected Result

leakage below 0.5 L/min and flow 2.3–2.5

Scenario 7: Piston problem

Field Condition

fluid leakage 4.5 L/min; discharge 150 barg; vibration 7.8

Problem

Piston rubber or rod condition is degrading.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses leakage above 1 L/min.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when leakage below 0.5 L/min.

Expected Result

leakage below 0.5 L/min

Scenario 8: Pulsation dampener problem

Field Condition

dampener precharge 38 barg versus 62 barg reference; pressure swing 19 barg

Problem

Low precharge is reducing pulsation absorption.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses precharge below 50 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when precharge set to approved 60–65 barg and swing below 5.

Expected Result

precharge set to approved 60–65 barg and swing below 5

Scenario 9: Power-end bearing problem

Field Condition

bearing 92 °C from 60–70; vibration 8.7; oil level below minimum

Problem

Bearing distress is developing in the power end.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses temperature above 85 °C.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when bearing 60–70 °C and vibration 3–5.

Expected Result

bearing 60–70 °C and vibration 3–5

Scenario 10: Lubrication problem

Field Condition

crankcase oil 28 °C; pressure 0.6 barg from 1.5–2.2; bearing 84 °C

Problem

Low oil pressure is reducing lubrication.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses pressure below 1.2 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.

3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when pressure 1.5–2.2 barg and bearing 60–70 °C.

Expected Result

pressure 1.5–2.2 barg and bearing 60–70 °C

Scenario 11: Discharge pressure abnormality

Field Condition

discharge 218 barg from 170–190; flow 1.7 m^3/min; current 410 A from 340–370

Problem

A blocked discharge or closed valve is overloading the drive.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses pressure above 205 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when pressure 170–190 barg and current 340–370 A.

Expected Result

pressure 170–190 barg and current 340–370 A

Scenario 12: Drive overload

Field Condition

motor current 445 A; 92 strokes/min; discharge 195 barg; vibration 6.5

Problem

High load or mechanical drag is causing drive overload.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses current above 420 A.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when current 340–370 A at approved stroke rate.

Expected Result

current 340–370 A at approved stroke rate

Scenario 13: Loose mounting

Field Condition

base movement 1.8 mm peak-to-peak; vibration 9.0; bolt torque loss 18 %

Problem

Loose skid/mounting is amplifying vibration.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses movement above 0.5 mm.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when movement below 0.5 mm and vibration 3–5.

Expected Result

movement below 0.5 mm and vibration 3–5

Scenario 14: Pipe/hose movement

Field Condition

discharge hose vibration 14 mm/s RMS; pump 5.2; pressure pulsation 11 barg

Problem

Inadequate restraint or damaged hose is transferring pulsation.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses hose vibration above 8 mm/s RMS.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when hose vibration below 4 mm/s RMS.

Expected Result

hose vibration below 4 mm/s RMS

Scenario 15: Vibration after liner replacement

Field Condition

vibration 8.3 mm/s RMS after replacement versus 3.1 before; 95 strokes/min

Problem

Incorrect liner fit, clearance, valve seating, or alignment is likely.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses increase above 3 mm/s RMS.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when vibration 3–5 mm/s RMS.

Expected Result

vibration 3–5 mm/s RMS